

RDS - Draft Report

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Proposed ADS: Home Credit Default Risk

Competition Host: Home Credit Group

Link: <https://www.kaggle.com/competitions/home-credit-default-risk>

Solution: <https://github.com/abhishekdbihani/Home-Credit-Default-Risk-Recognition>

We selected this ADS, because home loans are crucial to economic mobility, and whether a person is approved or denied can dramatically change the trajectory of their lives. It's therefore important to promote both group fairness and individual fairness, so a marginalized group isn't left behind, and everyone who deserves a loan gets it. Since there is historical bias in this field we want to ensure that the models we create are fair and uphold the principles we learned in class.

Background

The ADS that we plan to analyze in this project is a submission into Home Credit Group's Home Credit Default Risk Kaggle competition. Home Credit Group is a lender that strives to provide loans to underserved populations. The goal of the Kaggle competition is to develop a classifier tool that can assist Home Credit Group in their decision-making process for providing loans. The model we decided to audit classifies applicants as high-risk of default vs low. We decided to research the fairness of the model when it comes to gender. Historically, men tend to default more on their loans than women, so we'd like to see how the model handles these differences in base rates.

Credit history is an important factor assessed by lenders when determining the trustworthiness of prospective borrowers. For many, credit serves as a lifeline when money is not readily available. Through this ADS, the submitter hopes to utilize historical data to develop a tool that can be used for future credit lending decision-making. One of the major trade-offs that this model could yield is an inclusion of potentially high-risk borrowers. This may introduce a higher-likelihood of defaults on loans. This conflicts with the goal of providing credit to underserved groups.

Input and output

The data used was sourced from Home Credit Group who ran a Kaggle competition. The data collected was that of previous applicants (historical data). They extensively represent applicant information. The data files contain various data points that may be related to an applicant's credit worthiness and default risk such as financial profiles, employment history, and past behavior.

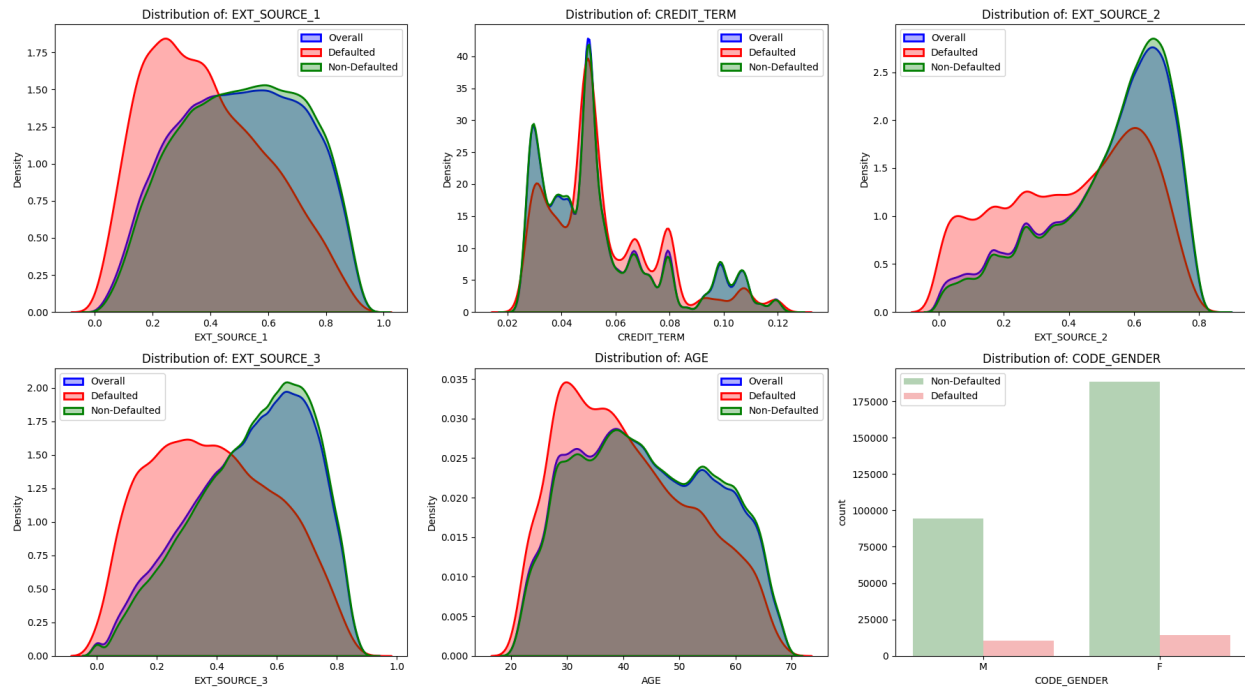


Figure 1

There are 1050 features used as inputs in the model and 307,511 entries within the data files, however, for explainability, we have decided to narrow that down to the 6 most important features – EXT_SOURCE_1 (float64), CREDIT_TERM (float64), EXT_SOURCE_3 (float64), EXT_SOURCE_2 (float64), AGE (float64), and CODE_GENDER (object). The first 5 of these features had the highest importance scores of all the features within the data files. The last variable – CODE_GENDER – is important because it is our sensitive feature as we are trying to understand how gender plays a role in credit approval and default risk. Amongst the 307,511 entries, only 134,133 have a valid EXT_SOURCE_1 value, 307,499 have a valid CREDIT_TERM value, 306,851 have a valid EXT_SOURCE 2 value, 246,546 have a valid EXT_SOURCE_3 value, all 307,511 have a valid AGE value, and 307,509 have a valid CODE_GENDER value.

The EXT_SOURCE features refer to normalized scores representing an applicant's creditworthiness as calculated by third-party providers. Thus, it is unshocking that they hold such great importance relative to the other features. CREDIT_TERM refers to the duration or length of the loan. AGE is in years, and is self-explanatory – older applicants may have more extensive credit history than younger applicants. Lastly, CODE_GENDER is a binary variable representing the gender of the applicant (either male or female). It is important to note that these data files contain a large class imbalance. The dominant class is non-defaulters who account for 92% of the data. The other 8% represent defaulters. This can create challenges because the model is incentivized to predict most as non-defaulters to maximize accuracy.

By observing the distributions among these inputs (Figure 1), we can see that defaulted loans tend to have lower external scores than the distribution of loans that weren't defaulted on.

Non-defaulted loans tended to follow the overall distribution. We observe that young borrowers in their 20s to 30s tend to default on loans than older borrowers. In our sensitive feature, gender, we observe that males have a higher rate of defaulting on loans than females. By analyzing the correlation matrix of these features (Figure 2) we can see that overall the features are uncorrelated with the exception of age and the first external source score which has a positive correlation of 0.6 and the external source and gender which has a positive correlation of 0.31. These low correlations are most likely due to how the features were selected.

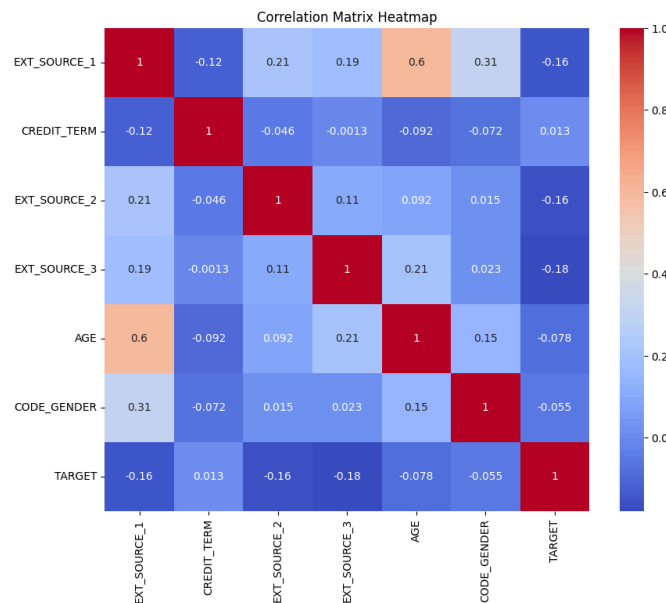


Figure 2

The output of the system is denoted as the variable called TARGET. The model provides probabilities of default, but given that this is a binary classifier, the prediction yields either 0 or 1. 0 represents a low default risk and 1 represents a high default risk. The threshold for this is set at 0.5. Meaning that anyone below this threshold would be labeled 0 and anyone above would be labeled 1.

Preliminary Analysis

For our preliminary analysis we decided to calculate the utility and fairness metrics of the ADS with respect to our sensitive feature gender. Overall, the model was accurate with an accuracy of 72.63% and a recall of 69.08% indicating that the model correctly identified the defaulted loans well (Figure 3). However, precision was very low at 19.23% indicating that most of the positive classifications were incorrect. The false positive rate was 30.91% and the false negative rate was 27.05% indicating poor performance. When we compare the performance among men and women we can see that the model performs quite differently. The model's accuracy among women was 76.56% while among men it was 65.09%. Precision was similar with 19.22% and 17.86% for women and men respectively. Recall was better for men at 76.02% and 63.89% for women. Men had a higher FPR at 36.13% and 22.46% for women. The FNR among women was 36.10% and 23.97% among men. This indicates that men are more likely to

be falsely classified to default on a loan while women are more likely to be falsely classified to not default.

Looking at the fairness metrics we can see that the FPR difference and FNR difference are about the same at 13.67% and 12.12% respectively. The selection rate difference of 14.78% means men are more likely to be classified as defaulting on a loan than women. The demographic parity ratio of 0.63 indicates the same thing. The equalized odds ratio was low as well at 0.62 indicating that there is a major fairness issue when it comes to false selections.

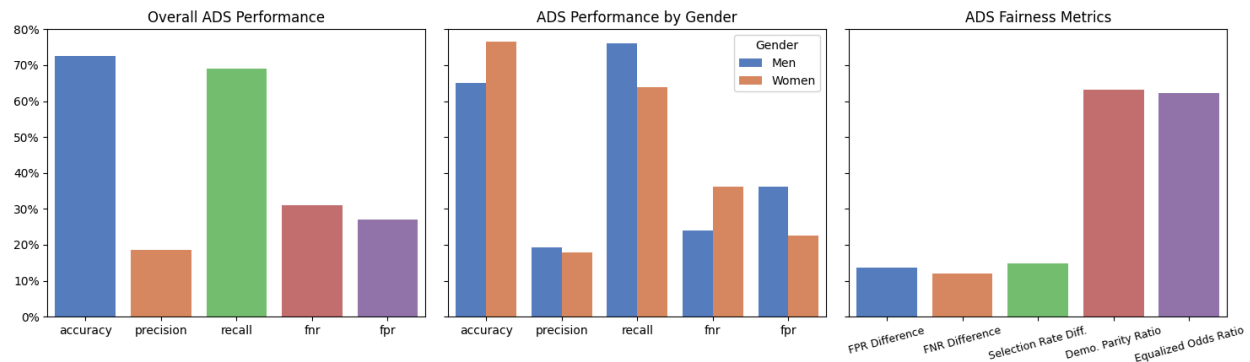


Figure 3

Next steps:

- Moving forward we would like to assess the fairness of the model with greater detail. Given that this model has meaningful real-world implications for numerous stakeholders, it is important that the model finds a balance between utility and fairness.
- We plan to do a detailed overlook of the solution, assessing how it handled the data cleaning and pre-processing. We'd like to know how nulls were handled and how new features were created.
- The author ultimately selects 1050 features to train the model, we want to understand how the author decided which features to use as inputs.
- We also want to see how the solution handles the core issue of the class imbalance. Poor handling of the class imbalance could lead to meaningless conclusions, thus it is pivotal that this step is done accordingly. From our preliminary analysis we see the solution uses an undersampling technique to address this class imbalance.
- We observe that the solution tested a variety of different models. We'd like to learn how the author evaluated these models to select the LightGBM model as the best. Furthermore we'd like to understand the optimization methods the author took to find the hyperparameters of the model.
- To further our understanding of fairness among the sensitive features, we plan to utilize SHAP values to understand the marginal contribution of model inputs.
- We will also implement PDPs to visualize features effects further.